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HI633

Does Joint Bracing Hinder Natural Mobility or Force Better Stride Consistency?

Hypothesis: While wearing a brace will predictably hinder mobility by shrinking a person's overall range of motion, it will simultaneously help them by acting as a mechanical stabilizer, causing their consecutive strides to become significantly more uniform and consistent compared to walking unbraced.

GitHub Repository: https://github.com/kafeiaulait-gif/HI-633-Final-Project

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Purpose of the Project

The purpose of this project is to investigate how mechanical joint restrictions alter baseline human locomotion across a sample of healthy subjects. Because lower-limb braces are traditionally designed to restrict movement, this study aims to evaluate if these devices serve strictly as a hindrance, or if they simultaneously function as dynamic stabilizers that enforce movement regularity across the entire lower body.

What the Project Investigates, Analyzes, and Predicts:

- Investigates: The tension between physical restriction and stabilization across the lower extremities. It explores how a restraint applied to a single location impacts overall walking harmony across multiple connected joints.
- Analyzes: The continuous joint angles (measured in degrees) across a series of complete, normalized stride cycles (0% to 100%). Specifically, the analysis tracks three distinct joints—the ankle, knee, and hip—bilaterally (on both the left and right legs). These measurements are evaluated under one of three conditions:
    - Condition 1: Unbraced (Normal baseline walking)
    - Condition 2: Knee brace applied to the left/right leg
    - Condition 3: Ankle brace applied to the left/right leg
- Predicts/Tests: The core hypothesis that while a unilateral restriction will predictably reduce absolute mobility (Range of Motion) in the targeted joint, it will simultaneously act as a structural anchor that forces consecutive stride cycles to become highly uniform and consistent across all three joints, thereby improving mobility and stability.

Dataset Description

The Multivariate Gait Data dataset is sourced from the UCI Machine Learning Repository at:

https://archive.ics.uci.edu/dataset/760/multivariate+gait+data

Dataset Overview

The dataset contains 181,800 instances and 7 features detailing the biomechanical analysis of human locomotion. It is a sequential, multivariate time-series dataset that captures continuous joint angles (measured in degrees) for the ankle, knee, and hip across both legs (bilateral).

Key Information Contained

- Subjects: Data collected from 10 healthy individuals.
- Conditions: 3 walking settings on a treadmill: normal (unbraced), knee braced (right knee), and ankle braced (right ankle).
- Replications: Each setting tracks 10 consecutive gait cycles starting and ending at heel-strike.
- Time Alignment: Each gait cycle is normalized into 101 time points representing 0% to 100% of the stride sequence.

Planned Analysis

To explore and analyze the dataset, the plan relies on a structured machine-learning pipeline utilizing Python and standard foundational algorithms. The analysis is broken down into four key stages:

1. Exploratory Data Analysis (EDA)
    - Data Visualization: Plotting continuous joint angles across the 0% to 100% gait cycle using matplotlib to visually inspect differences between the three walking conditions.
    - Anomaly Detection: Inspecting the data structure for missing values, out-of-range degrees, or incomplete stride replications.
2. Feature Engineering & Dimensionality Reduction
    - Summary Statistic Extraction: Transforming raw, complex time-series sequences into simplified numeric features by calculating Maximum Angle, Minimum Angle, and Range of Motion (ROM) for each of the three joints (ankle, knee, hip).
    - Data Reshaping: Pivot-transforming the long gait cycle rows into a structured wide format where each individual stride replication represents a single sample observation.
3. Predictive Machine-Learning Modeling
    - Data Partitioning: Splitting the processed dataset into independent training (70%) and testing (30%) subsets using stratified sampling to maintain class balance.
    - Supervised Classification: Training a Random Forest Classifier or Decision Tree algorithm to predict the target label—the specific bracing condition (1, 2, or 3)—based on the extracted joint mobility features.
4. Model Evaluation & Performance Metrics
    - Accuracy Assessment: Generating a classification report to measure precision, recall, and F1-score for each condition type.
    - Feature Importance Analysis: Extracting model weightings to determine which specific joint angles (e.g., right knee vs. right hip) provide the strongest predictive power when detecting physical restrictions.

References

Google. (2026, Oct. 20). Synthetic Healthcare Dataset Search [Generative AI chat]. Google Gemini. https://share.google/aimode/A4JO9lLqBPqwZgdps

Helwig, N. & Hsiao-Wecksler, E. (2016). Multivariate Gait Data [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5861T.